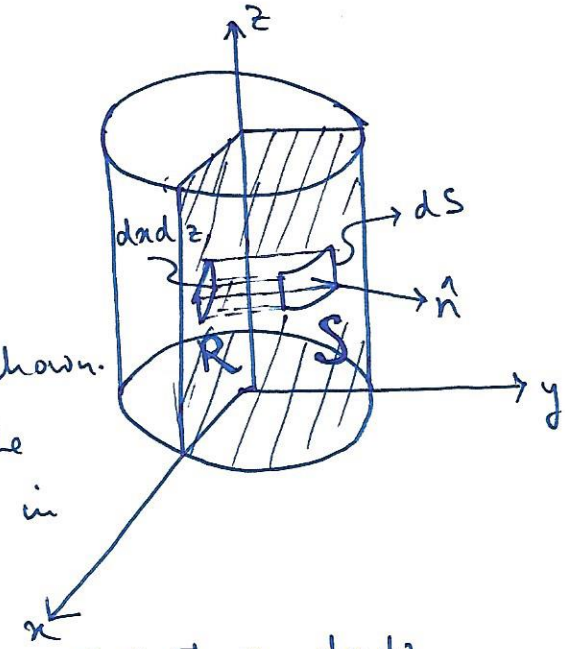
1. Evaluate $\iint_S \vec{A}\cdot\hat{n}\,dS$ where $\vec{A} = z\hat{i} + x\hat{j} - 3y^2z\hat{k}$ and S is the surface of the cylinder $x^2+y^2=16$ included in the first octant between $z=0$ and $z=5$. ①

The region of integration is the shaded region (S) which is the curved surface of the cylinder. Consider an elementary area 'dS' in S having unit normal $\hat{n}$ as shown. Project the area element dS in the x-z plane – this element is dx dy in the region R (x-z plane).



In such a case, $\iint_S \vec{A}\cdot\hat{n}\,dS = \iint_R \vec{A}\cdot\hat{n}\,\dfrac{dx\,dz}{|\hat{n}\cdot\hat{j}|}$ if projection is in x-z plane. [see lecture slides]

Now we know that if $\phi(x,y,z)=c$, then $\vec{\nabla}\phi$ represents the direction normal to the surface.

$\vec{n} = \vec{\nabla}(x^2+y^2-16) = 2x\hat{i} + 2y\hat{j}$; $\hat{n} = \dfrac{2x\hat{i}+2y\hat{j}}{2\sqrt{x^2+y^2}}$

$= \dfrac{x\hat{i}+y\hat{j}}{4}$ $[\because x^2+y^2=16]$

$\vec{A}\cdot\hat{n} = \frac{1}{4}(xz+xy)$

$\hat{n}\cdot\hat{j} = y/4$. Then the surface integral equals,

$$\iint_R \left(\frac{xz+xy}{y}\right)dx\,dz = \int_{z=0}^{5}\int_{x=0}^{4}\left(\frac{xz}{\sqrt{16-x^2}} + x\right)dx\,dz = I_1 + I_2$$

$$I_1 = \int_{z=0}^{5}\int_{x=0}^{4} \frac{xz}{\sqrt{16-x^2}}\,dx\,dz$$

$$= \int_{z=0}^{5} z\,dz \int_{x=0}^{4} \frac{x\,dx}{\sqrt{16-x^2}} = \frac{5^2}{2} \times -\int_0^4 \frac{d(16-x^2)}{\sqrt{16-x^2}}$$

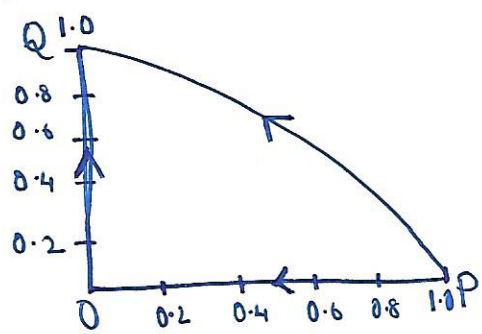
$$= \frac{25}{2} \cdot 2\sqrt{16-x^2}\Big|_0^4$$

$$= \frac{25}{4} \times 8 = 50$$

$$I_2 = \iint x\,dx\,dz = \int_{x=0}^{4} x\,dx \int_0^5 dz = 5 \cdot \frac{4^2}{2} = 40$$

$$\therefore\ I = I_1 + I_2 = 50 + 40 = 90$$

2. A particle is moved under the influence of a force $\vec{F}(x,y) = -ax\,\hat{i} - by\,\hat{j}$ from point P (1,0) to point Q (0,1) in two ways: (i) along the axes (P→O→Q) (ii) along the quarter circle (P→Q). Here 'a' and 'b' are constants with appropriate units



(a) Calculate the work done by the force in both cases

(b) Do the two results match? What does it tell you about the nature of force?

(c) Can $F(x,y)$ be expressed as the gradient of a scalar function? If yes, try to obtain it.

(d) Evaluate curl of $\vec{F}(x,y)$. Is the answer consistent with above results

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(a) Work done is given by the line integral of the force along the given path:

$$W = \int \vec{F}\cdot d\vec{r} = \int F_x\,dx + \int F_y\,dy$$

$$= -\int ax\,dx - \int by\,dy$$

<u>Along PO:</u> $y = 0$, $W = -\int_1^0 ax\,dx = \frac{a}{2}$

$x: 1 \to 0$

<u>Along OQ:</u> $x = 0$, $y: 0 \to 1$, $W = -\int_0^1 by\,dy = -\frac{b}{2}$

$$W(PO + OQ) = \frac{a}{2} - \frac{b}{2} = \int_{PO+OQ} \vec{F}\cdot d\vec{r}$$

<u>Along curved surface PQ:</u>

Equation of curved surface PQ is $x^2 + y^2 = 1$

$$W = \int F_x\,dx + \int F_y\,dy = -\int ax\,dx - \int by\,dy$$

Along PQ; $y = \sqrt{1-x^2}$

$$dy = -\frac{1 \cdot \cancel{2}x}{\cancel{2}\sqrt{1-x^2}} = -\frac{x}{\sqrt{1-x^2}}$$ [as x increases, y decreases in first quadrant]

'x' varies from 0 to 1 along PQ

$$W = -\int_{x=1}^{0} a x \, dx + \int_{x=1}^{0} b\sqrt{1-x^2}\,\frac{x}{\sqrt{1-x^2}}\,dx$$

$$= \left(\frac{a}{2} - \frac{b}{2}\right) \text{ along } PQ$$

(b) Work done along the two paths are same. This is an indication of conservative force.
If line integral of a vector function is same along different paths connecting same two points, then the vector function is conservative.

(c) A conservative vector function can be expressed as the gradient of a scalar function, that is here $\vec{F} = \vec{\nabla}\phi(x,y)$. $\phi(x,y)$ can be deduced mathematically.

$$-ax\,\hat{i} - by\,\hat{j} = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y}$$

$$\frac{\partial\phi(x,y)}{\partial x} = -ax$$

$$\phi(x,y) = -a\frac{x^2}{2} + f(y)$$

[partial deriv. w.r.t. x - so we can have a func. of y]

$$\frac{\partial\phi(x,y)}{\partial y} = -by$$

$$\phi(x,y) = -\frac{b}{2}y^2 + g(x)$$

Comparing, $f(y) = -b\frac{y^2}{2}$, $g(x) = -a\frac{x^2}{2}$

$$\boxed{\phi(x,y) = -a\frac{x^2}{2} - b\frac{y^2}{2}}$$

. Note that $U = -\phi = \frac{a}{2}x^2 + \frac{b}{2}y^2$ is the potential energy of a 2D harmonic oscillator and $\vec{F} = -\vec{\nabla}U = -ax\,\hat{i} - by\,\hat{j}$ is a 2-D spring force (Hooke's law).

(d) $$\vec{\nabla} \times \vec{F}(x,y) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -ax & -by & 0 \end{vmatrix}$$

$$= \hat{i}\left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(by)\right] + \hat{j}\left[\frac{\partial}{\partial z}(-ax) - \frac{\partial}{\partial x}(0)\right] + \hat{k}\left[\frac{\partial}{\partial x}(-by) - \frac{\partial}{\partial y}(-ax)\right]$$

$$= 0$$

This is consistent with the results in (a), (b), (c). We know that $\vec{\nabla} \times (\vec{\nabla}\phi)$ is also zero for any scalar function ϕ. So $\vec{F}$ must be equal to $\vec{\nabla}\phi$.

This has been proved in (c) where ϕ has been found explicitly. Equivalently, $\vec{\nabla} \times \vec{V}_c = 0$ for a conservative vector field V_c. Here $\vec{V}_c \equiv \vec{F}$ and we already proved in (b) that it is conservative.

3. Calculate the Laplacian of the following:

(a) $g(x,y,z) = \frac{1}{12}x^4 - \ln\left(\frac{y}{z}\right)$

$$g(x,y,z) = \frac{1}{12}x^4 - \ln y + \ln z$$

$$\nabla^2 g(x,y,z) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\left\{\left(\frac{1}{12}x^4 - \ln y + \ln z\right)\right\}$$

$$= \frac{\partial^2}{\partial x^2}\left(\frac{1}{12}x^4 - \ln y + \ln z\right) + \frac{\partial^2}{\partial y^2}\left(\frac{1}{12}x^4 - \ln y + \ln z\right)$$

$$+ \frac{\partial^2}{\partial z^2}\left(\frac{1}{12}x^4 - \ln y + \ln z\right)$$

$$= \frac{12x^2}{12} + \frac{1}{y^2} - \frac{1}{z^2} = x^2 + \frac{1}{y^2} - \frac{1}{z^2}$$

(b) $\vec{V}(x,y,z) = e^{-xyz}\hat{i} + z^2\hat{j} + xz\hat{k}$

$$\nabla^2\vec{V}(x,y,z) = \nabla^2(\hat{i}V_x + \hat{j}V_y + \hat{k}V_z) = \hat{i}\nabla^2 V_x + \hat{j}\nabla^2 V_y + \hat{k}\nabla^2 V_z$$

$$= \hat{i}\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)e^{-xyz}\right] + \hat{j}\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)(z^2)\right]$$

$$+ \hat{k}\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)(xz)\right]$$

$$\cancel{= \hat{i}\left[(x^2+y^2+z^2)e^{-xyz}\right] + \hat{j}\times 2}$$

$$= \hat{i}\left[(y^2z^2 + x^2z^2 + x^2y^2)e^{-xyz}\right] + 2\hat{j}$$